Def° $\mathcal{D}_X$. sheaf of non-comm. alg on X.

. sheaf of $\mathcal{O}_X \otimes \mathcal{O}_X$ -modules

- generated over $\mathcal{O}_X \otimes \mathcal{O}_X$ by T_X

- filtration by order, $gr(\mathcal{D}_X) = \mathcal{O}_{T^*_X}$ (as sheaf on X)

Example: $X = \mathbb{A}^1$ choose ∂ generator for T_X (dual to dx)

$\partial = \frac{d}{dx}$

$$\mathcal{D}_X(X) = \frac{k\langle x, \partial\rangle}{[\partial, x] = 1} \qquad [D, x] = D(x)$$

$\downarrow$

∂ is derivation $\frac{d}{dx}$

$$\mathcal{U} = \mathbb{G}_m \quad \mathcal{D}_X(\mathcal{U}) = \frac{k\langle x, x^{-1}, \partial\rangle}{\begin{array}{l}[\partial, x] = 1 \\ [\partial, x^{-1}] = -x^{-2}\end{array}}$$

$\frac{d}{dx}(x^{-1}) = -x^{-2}$.

Ex: $X = \mathbb{P}^1$ Choose coords x, y on affins, $xy = 1$

$$\mathcal{D}_X(\mathcal{U}_x) = \frac{k\langle x, \partial_x\rangle}{[\partial_x, x] = 1}$$

$$\mathcal{D}_X(\mathcal{U}_y) = \frac{k\langle y, \partial_y\rangle}{[\partial_y, y] = 1}$$

glue: $\partial_x = -y^2 \partial_y$ quantized T_X = has global sections.

chain rule

$$\frac{d}{dx} = \frac{dy}{dx} \cdot \frac{d}{dy} = -y^2$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2} = y^2$$

Exercise. Show that $\mathcal{D}_X$ is globally generated.

Def: The category of left D-modules on X $\quad D^l(X)$

is the category of $\mathcal{O}_X$-quasicoherent sheaves of left $\mathcal{D}_X$-modules

right D-modules $\quad D^r(X)$

Sometime / usually ... $D(X)$ context, left

Remarks ① Because $\mathcal{D}_X$ quasicoherent, "$\mathcal{O}_X$-quasicoh $\iff$ $\mathcal{D}_X$-quasicoh"

i.e. locally any D-mod $\mathcal{M}$ has pres.

X smooth always

$$\mathcal{M} = \operatorname{coker}\Big(\bigoplus_{j\in J} \mathcal{D}_X|_{\mathcal{U}} \longrightarrow \bigoplus_{i\in I} \mathcal{D}_X|_{\mathcal{U}}\Big)$$

② **Def:** $\mathcal{M}$ is coherent if J, I finite. $\iff$ I finite $\quad D_c(X)$ $\Big\}$ 1.4.9 HTT

Singular support:

Def: A filtration on $\mathcal{M}$ (D-module), compatible w/ order filt on $\mathcal{D}_X$ is good if $gr(\mathcal{M})$ is coherent as $gr(\mathcal{D}_X) \cong \operatorname{Sym}_X T_X$ module.

$(F_i\mathcal{D}_X)\cdot(F_j\mathcal{M}) \subseteq F_{i+j}\mathcal{M}$

$\iff$ $gr(\mathcal{M})$ coherent sheaf on $T_X^* = \operatorname{Spec}_X \operatorname{Sym}_X T_X$

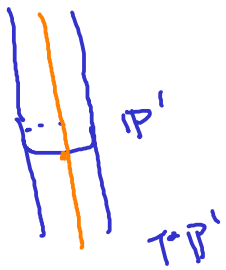
Define: singular support of $\mathcal{M}$

$$SS(\mathcal{M}) = \operatorname{supp}(gr(\mathcal{M})) \subseteq T_X^*$$

for some good filtration

Prop: · If $\mathcal{M}$ is coh, has a good filtration.
· SS independent of choice of good filtration.

Observations: $SS(\mathcal{M}) \subseteq T^*_X$ · conical
$\circlearrowleft$ · closed
$\mathbb{G}_m$ on · image is X = restriction to X
fibers = usual support.



Thm: If $\mathcal{M}$ is locally $\mathcal{O}_X$-free, it's a flat connection.
$\iff SS(\mathcal{M}) = X$ zero section.

SS measures failure of $\mathcal{M}$ to be a flat connection.

Ex: $X = \mathbb{A}^1$, $U = \mathbb{G}_m$ coords: ∂, x $\partial \xrightarrow{gr} \xi$

① $D_X \curvearrowright \mathcal{O}_X$ $D_X \subseteq End_k(\mathcal{O}_X)$

$D_X \xrightarrow{\varphi} \mathcal{O}_X$ $\ker(\varphi) \supseteq D_X \cdot \partial$
$D \longmapsto D \cdot 1$

Write:
$D = \sum f_n(x) \partial^n$
$D \cdot 1 = f_0(x) = 0.$
$\iff D \in D_X \cdot \partial.$

Singular support: trivial filtration is good.

$0 \subseteq \mathcal{O}_X \subseteq \mathcal{O}_X$
$gr \{ \quad \partial$
$0 \oplus \mathcal{O}_X \oplus 0$
ξ

$gr(\mathcal{O}_X) = k[x, \xi]$ $SS(\mathcal{O}_X) = X$ zero section.
$\xi = 0$ ← cotangent ∂

$x^m \partial_x^n$ ② D_x acts on $\mathcal{O}_M : k[x, x^{-1}]$

1 ∂ is not a good

never get $\frac{1}{x}$

① $\partial^n \to D_x^{\leq n}$?

Claim:

$D_x \longrightarrow \mathcal{O}_M$

$D \longmapsto D \cdot \frac{1}{x}$ generator

$\ker = D_x \cdot \partial x$

look for diff eq $\frac{1}{x}$ satisfies

$f = \frac{1}{x}$ is solution to $\frac{d}{dx} x f = 0$.

$\frac{d}{dx} \frac{1}{x^2} \xleftarrow{\partial} \frac{1}{x} \xrightarrow{x} 1 \xrightarrow{x} x^2 \to \cdots$

$0 \xleftarrow{\partial}$

$D = \sum f_n(x) \partial_x^n + \sum g_n \partial^n$?

Exercise: · Compute SS

· Do the same for $D_x / D_x \cdot x$

· $M_\lambda = D_x / D_x (x\partial - \lambda)$

Which of these M_λ are isomorphic to each other?

Exercise: $gr(M)$

good filtration:

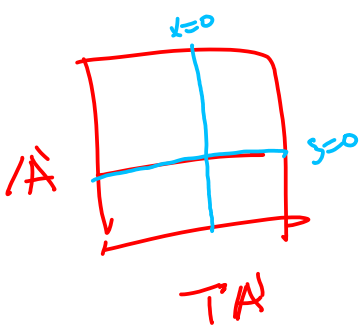
$\mathcal{O}_x \subseteq \frac{1}{x} \mathcal{O}_x \subseteq \frac{1}{x^2} \mathcal{O}_x \subseteq \cdots$

∂ ∂

gr

$\frac{k[x,\xi]}{x\xi}$

start here



supp

$k[x] \oplus k[\xi] = \mathcal{O}_x \oplus k \cdot \frac{1}{x} \oplus k \cdot \frac{1}{x^2} \oplus \cdots$

$\xi = 0$ ξ

$\xi x = 0$

$\xi \cdot \frac{1}{x} = \frac{1}{x^2}$

$\frac{1}{x} \mathcal{O}_x \oplus k \frac{1}{x^2} \oplus \cdots$